Introduction to Basic and Advanced Statistical Modelling

Using R (solutions to exercises)

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Table of contents

Reading data

Practice 4

```r
territories <- read_csv("data_MHB/data-territories.csv")
```

```
Rows: 1152306 Columns: 10
-- Column specification ------------------------------------------------------
Delimiter: ","
chr (2): nameeg, namelt
dbl (8): species_id, euring_id, samplearea_id, year, territories_visit_1, territories_...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Wrangling data

Practice 5a

Use `select()` and/or `filter()` and/or `slice()` and/or `arrange()` to determine:

1. the largest number of recorded species across all years

```
species_count |>
    arrange(desc(n_recorded_species)) |>
    slice(1) |>
    select(n_recorded_species)
```

or

```
species_count |>
    filter(n_recorded_species == max(n_recorded_species)) |>
    select(n_recorded_species)
```

Practice 5b

Use `select()`, `filter()`, `slice()` and/or `arrange()` to determine:

2. the largest number of recorded species in 2025

```
species_count |>
    filter(year == 2025) |>
    filter(n_recorded_species == max(n_recorded_species)) |>
    select(n_recorded_species)
```

but NOT this:

```
species_count |>
    filter(year == 2025, n_recorded_species == max(n_recorded_species)) |>
    select(n_recorded_species)
```

Do you understand why?

Practice 6a

Use the functions we have seen to obtain:

1. the range of median elevation according to the number of visits

```
species_count |>
    summarise(median_elevation_min = min(median_elevation),
              median_elevation_max = max(median_elevation),
              .by = n_visits)
```

```
# A tibble: 2 x 3
  n_visits median_elevation_min median_elevation_max
     <dbl>                <dbl>                <dbl>
1        3                  250                 2150
2        2                 1450                 2750
```

Practice 6b

Use the functions we have seen to obtain:

2. the number of observations for *Sturnus vulgaris* for each year

```
territories |>
    filter(namelt == "Sturnus vulgaris") |>
    summarise(territories_sum = sum(territories_total, na.rm = TRUE), # NAs present
              .by = year)
```

```
# A tibble: 27 x 2
   year territories_sum
  <dbl>           <dbl>
1  1999             691
2  2000             920
3  2001             911
4  2002             887
# i 23 more rows
```

Practice 6c

Use the functions we have seen to obtain:

3. the 3 most recorded species for each year

```
territories |>
    summarise(territories_sum = sum(territories_total),
              .by = c(year, nameeg, namelt)) |> # aggregate across samplearea_id
    slice_max(territories_sum, n = 3, by = year) |> # pick top 3 per year
    select(year, nameeg, namelt, territories_sum)
```

```
# A tibble: 81 x 4
   year nameeg             namelt            territories_sum
  <dbl> <chr>              <chr>                       <dbl>
1  1999 Eurasian Chaffinch Fringilla coelebs            5775
2  1999 Common Blackbird   Turdus merula                3132
3  1999 Coal Tit           Periparus ater               3108
4  2000 Eurasian Chaffinch Fringilla coelebs            6695
# i 77 more rows
```

Practice 7

Try to use `pivot_longer()` to get back to the original long format of the data from the output of the following code:

```
territories |>
    filter(year == 2025) |>
    select(samplearea_id, nameeg, territories_total) |>
    pivot_wider(names_from = nameeg,
                values_from = territories_total) |>
    pivot_longer(-samplearea_id,
                 names_to = "nameeg",
                 values_to = "territories_total")
```

```
# A tibble: 42,282 x 3
  samplearea_id nameeg              territories_total
          <dbl> <chr>                           <dbl>
1           268 Little Grebe                        0
2           268 Great Crested Grebe                 0
3           268 Grey Heron                          0
4           268 Little Bittern                      0
# i 42,278 more rows
```

Wrangling spatial data

Practice 8a

Use the functions we have seen to obtain:

1. the sampled area (sensu convex hull) across years

```
species_count_sf |>
    summarise(geometry = st_combine(geometry), .by = year) |>
    mutate(hull = st_convex_hull(geometry),
           area = st_area(hull),
           area_km2 = units::set_units(area, "km^2")) |>
    select(-hull) |>
    st_drop_geometry()
```

```
# A tibble: 27 x 3
   year         area area_km2
* <dbl>        [m^2]   [km^2]
1  1999 48899915383.   48900.
2  2000 48995663682.   48996.
3  2001 48229824468.   48230.
4  2002 48995663682.   48996.
# i 23 more rows
```

Practice 8b (1)

Use the functions we have seen to obtain:

2. the IDs of the sampled areas which are most peripheral

Simple sorting by distance to centroid (but lacks threshold)

```
species_count_sf |>
  distinct(samplearea_id, geometry) |>
  mutate(centroid = st_centroid(st_combine(geometry)),
         dist = st_distance(centroid, geometry, by_element = TRUE)) |>
  arrange(desc(dist)) |>
  as_tibble() # (for slide only -> skip metadata clutter)
```

```
# A tibble: 267 x 4
  samplearea_id            geometry            centroid    dist
          <dbl>         <POINT [°]>         <POINT [°]>     [m]
1           269 (6.110458 46.21013) (8.234991 46.79076) 174955.
2           398 (10.40901 46.61099) (8.234991 46.79076) 166982.
3           271 (6.184874 46.35493) (8.234991 46.79076) 164026.
4           397 (10.34234 46.82876) (8.234991 46.79076) 160429.
# i 263 more rows
```

Practice 8b (2)

More useful, but more complex: intersection with concave hull outline

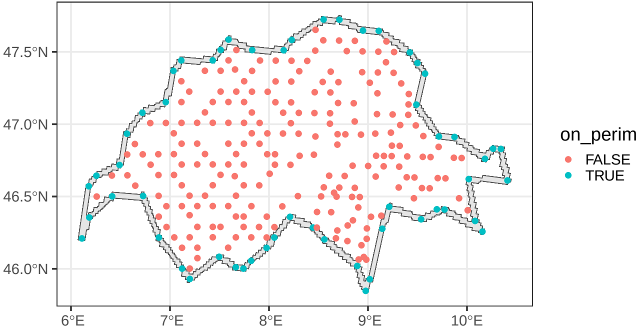
```
species_count_sf |>
  distinct(samplearea_id, .keep_all = TRUE) |>
  mutate(all_points = st_combine(geometry), # combine points into new geometry
         hull = st_concave_hull(all_points, ratio = 0.2), # create concave hull
         perimeter = st_boundary(hull), # extract outer line of the hull
         perimeter = st_buffer(perimeter, dist = 1000), # create buffer zone around it
         on_perim = st_intersects(perimeter, geometry, # test if points fall into buffer
                                  by_element = TRUE)) -> data_perimeter_sf

data_perimeter_sf |>
    filter(on_perim) |>
    pull(samplearea_id) # pull() extracts a column
```

```
 [1] 269 271 273 277 278 279 282 283 286 289 291 294 302 307 308 311 323 325 340 352 359
[22] 369 370 376 384 385 388 392 393 394 395 396 397 398 400 408 411 418 427 434 439 456
[43] 460 461 462 469 475 476 478 502 503 515 522 525 531
```

Practice 8b (2)

More useful, but more complex: intersection with concave hull outline

```
theme_set(theme_bw(base_size = 20)) # setting the theme for all plots
ggplot(data_perimeter_sf) +
  geom_sf(aes(geometry = perimeter)) +
  geom_sf(aes(colour = on_perim)) # geometry = geometry is implicit
```



Plotting data

Practice 10 (1)

- combine data wrangling and plotting to illustrate the change in species abundance across years for the top 6 most encountered species

```
territories |>
    summarise(territories_sum = sum(territories_total),
              .by = c(nameeg, namelt)) |>
    slice_max(territories_sum, n = 6) |>
    pull(namelt) -> top6sp

top6sp
```

```
[1] "Fringilla coelebs"       "Parus major"             "Troglodytes troglodytes"
[4] "Phylloscopus collybita"  "Turdus viscivorus"       "Columba palumbus"
```

Practice 10 (2)

- combine data wrangling and plotting to illustrate the change in species abundance across years for the top 6 most encountered species

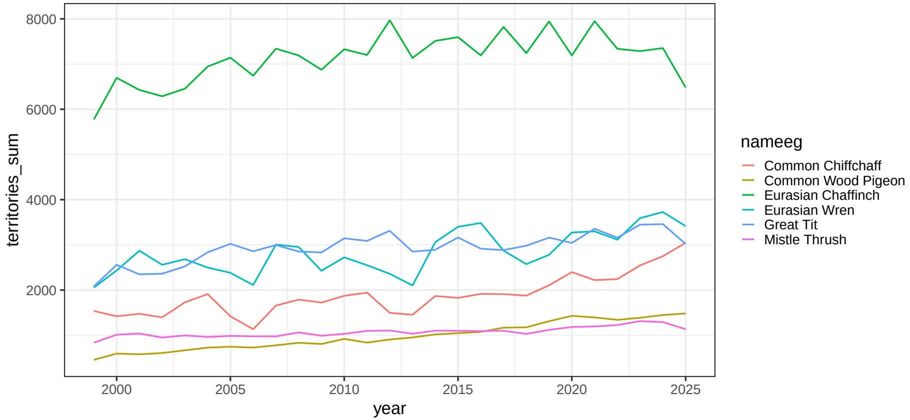
```
territories |>
    filter(namelt %in% top6sp) |>
    summarise(territories_sum = sum(territories_total),
              .by = c(nameeg, namelt, year)) -> top6_across_time

top6_across_time
```

```
# A tibble: 162 x 4
  nameeg             namelt             year territories_sum
  <chr>              <chr>             <dbl>           <dbl>
1 Common Wood Pigeon Columba palumbus   1999             457
2 Common Wood Pigeon Columba palumbus   2000             594
3 Common Wood Pigeon Columba palumbus   2001             578
4 Common Wood Pigeon Columba palumbus   2002             607
# i 158 more rows
```

Practice 10 (3a)

- combine data wrangling and plotting to illustrate the change in species abundance across years for the top 6 most encountered species

```r
ggplot(top6_across_time) +
  geom_line(aes(y = territories_sum, x = year, colour = nameeg))
```



Practice 10 (3b)

- combine data wrangling and plotting to illustrate the change in species abundance across years for the top 6 most encountered species

```
ggplot(top6_across_time) +
  geom_line(aes(y = territories_sum, x = year)) +
  facet_wrap(~ nameeg, scales = "free_y")
```